



A bidirectional interpretable compound-protein interaction prediction framework based on cross attention

Meng Wang^a, Jianmin Wang^b, Zhiwei Rong^c, Liuying Wang^a, Zhenyi Xu^a, Liuchao Zhang^a, Jia He^a, Shuang Li^a, Lei Cao^a, Yan Hou^c, Kang Li^{a,*}

^a School of Public Health, Harbin Medical University, Harbin, 150081, China

^b School of Integrative Biotechnology and Translational Medicine, Yonsei University, Incheon, 21983, Republic of Korea

^c School of Public Health, Peking University, Beijing, 100871, China

ARTICLE INFO

Keywords:

Compound-protein interaction
Cross-attention mechanism
Bidirectional interpretable deep learning model
Drug discovery

ABSTRACT

The identification of compound-protein interactions (CPIs) plays a vital role in drug discovery. However, the huge cost and labor-intensive nature in vitro and vivo experiments make it urgent for researchers to develop novel CPI prediction methods. Despite emerging deep learning methods have achieved promising performance in CPI prediction, they also face ongoing challenges: (i) providing bidirectional interpretability from both the chemical and biological perspective for the prediction results; (ii) comprehensively evaluating model generalization performance; (iii) demonstrating the practical applicability of these models. To overcome the challenges posed by current deep learning methods, we propose a cross multi-head attention oriented bidirectional interpretable CPI prediction model (CmhAttCPI). First, CmhAttCPI takes molecular graphs and protein sequences as inputs, utilizing the GCW module to learn atom features and the CNN module to learn residue features, respectively. Second, the model applies cross multi-head attention module to compute attention weights for atoms and residues. Finally, CmhAttCPI employs a fully connected neural network to predict scores for CPIs. We evaluated the performance of CmhAttCPI on balanced datasets and imbalanced datasets. The results consistently show that CmhAttCPI outperforms multiple state-of-the-art methods. We constructed three scenarios based on compound and protein clustering and comprehensively evaluated the model generalization ability within these scenarios. The results demonstrate that the generalization ability of CmhAttCPI surpasses that of other models. Besides, the visualizations of attention weights reveal that CmhAttCPI provides chemical and biological interpretation for CPI prediction. Moreover, case studies confirm the practical applicability of CmhAttCPI in discovering anticancer candidates.

1. Introduction

The identification of compound-protein interactions (CPIs) is an essential task in various applications of drug repurposing and prediction of possible side effects [1,2]. Though experimental assays of compound-protein binding affinity remain accurate, they are often hindered by high costs and time constraints [3]. Over the past decades, numerous computational models for CPI prediction have been proposed to narrow down the search scope of drug and protein candidates for reducing cost, enhancing efficiency of drug discovery and accelerating drug development [4].

In recent years, many deep learning methods have been developed to identify CPIs [5–7]. However, early models require to precomputed

fixed-rule drug fingerprints and protein descriptors, potentially inducing prediction bias due to missing structural information during the computation. With the rapid development of deep learning methods for drug development, many end-to-end algorithms have been developed for CPI research [8–13]. End-to-end models seamlessly integrate representations learned from raw data and operate in unified frameworks. Unlike early models, end-to-end approaches eliminate the need for predefined fingerprints and descriptors. DeepDTA [11] applies convolutional neural networks (CNNs) to learn low-dimensional deep representations from simplified molecular-input line-entry system (SMILES) strings [14] and protein sequences, respectively. Then, these representations are then concatenated and fed into a fully connected neural network (FCNN) for the calculation of compound-protein

* Corresponding author.

E-mail address: likang@ems.hrbmu.edu.cn (K. Li).

<https://doi.org/10.1016/j.combiomed.2024.108239>

Received 31 August 2023; Received in revised form 25 February 2024; Accepted 26 February 2024

Available online 2 March 2024

0010-4825/© 2024 Elsevier Ltd. All rights reserved.

affinities. Graph-CNN [12] learns representations of protein pockets from druggable protein binding sites, and then constructs two graph convolutional neural networks (GCNNs) for pocket graphs and 2D compound graphs to automatically extract features of CPIs. GrpahDTA [15] employs graph neural networks (GNNs) to extract drug representations and CNNs to capture protein representations, respectively. The drug and protein representations are concatenated and passed through a FCNN to estimate drug–target affinities. Despite these end-to-end frameworks have achieved promising performance in CPI prediction, their interpretability remains unsatisfactory due to a lack of detailed processing of local crucial features that contribute significantly to CPIs.

A group of end-to-end methods can capture local important features with attention mechanism [16] widely adopted in deep-learning community to increase the interpretability [8,17–19]. CPI-GNN can emphasize specific subsequences of proteins that are crucial for molecules [8]. However, regarding compounds, this study does not provide a detailed analysis of atom attention weights, specifically whether atoms with high attention weights interact with these subsequences. TransformerCPI utilizes the attention mechanism to capture various compound features corresponding to different proteins and provides a rough speculation on whether a new compound binds to the transmembrane or extracellular region of a GPCR protein [17]. Nevertheless, for a specific CPI, TransformerCPI does not simultaneously elucidate which residues and atoms are crucial for the interaction. CoaDTI employs a co-attention mechanism to measure the interaction on the drug side and the protein side synchronously, capturing crucial atoms and residues involved in the interaction [18]. But the high-attention residues are not associated to biological functions. ICAN investigates the actual focus of the attention mechanism by shifting the attention binding sites in proteins [19]. However, the detailed descriptions or analyses relating to compounds are not exhibited. Although existing end-to-end models with attention machine have demonstrated binding sites of compounds or proteins in a

few examples, they still face the challenge of providing bidirectional interpretation for CPI prediction results from both the chemical and biological perspective.

In addition, the current approaches also encounter the challenge of lacking comprehensive evaluation on model generalization ability with effective scenarios. In previous CPI studies [18,20,21], it is common to split datasets using unknown compound and unknown protein schemes to evaluate model generalization ability. In the unknown compound scheme, compounds from the test set are not present in the training set. Similarly, in the unknown protein scheme, proteins from the test set are not present in the training set. However, in the unknown compound scheme, most comparative studies suffer from compound series bias [22], leading to a potential overestimation of the performance of certain methods. The compound series bias arises from the generation of chemical compounds. Specifically, chemists commonly create new chemical scaffolds instead of individual compounds, and they develop new substances by incorporating various functional groups into these scaffolds [23]. This issue is also evident in the unknown protein scheme due to the similarities between proteins.

To address these challenges, we propose a novel cross multi-head attention oriented bidirectional interpretable deep learning model called CmhAttCPI (Fig. 1). The proposed model consists of four parts including graph convolutional networks with a graph warp unit (GCW) module, CNN module, cross multi-head attention module, and a fully connected neural network (FCNN). CmhAttCPI applies GCW module to extract atom features from molecular graphs and CNN module to extract residue features from protein sequences. In cross multi-head attention module, the extracted atom and residue features are fed into two multi-head scaled dot-product attentions to generate atom and residue attention weights, respectively. Then, the final atom and residue features are obtained by adding atom and residue attention weights to atom and residue features, respectively. Finally, with the final atom and residue

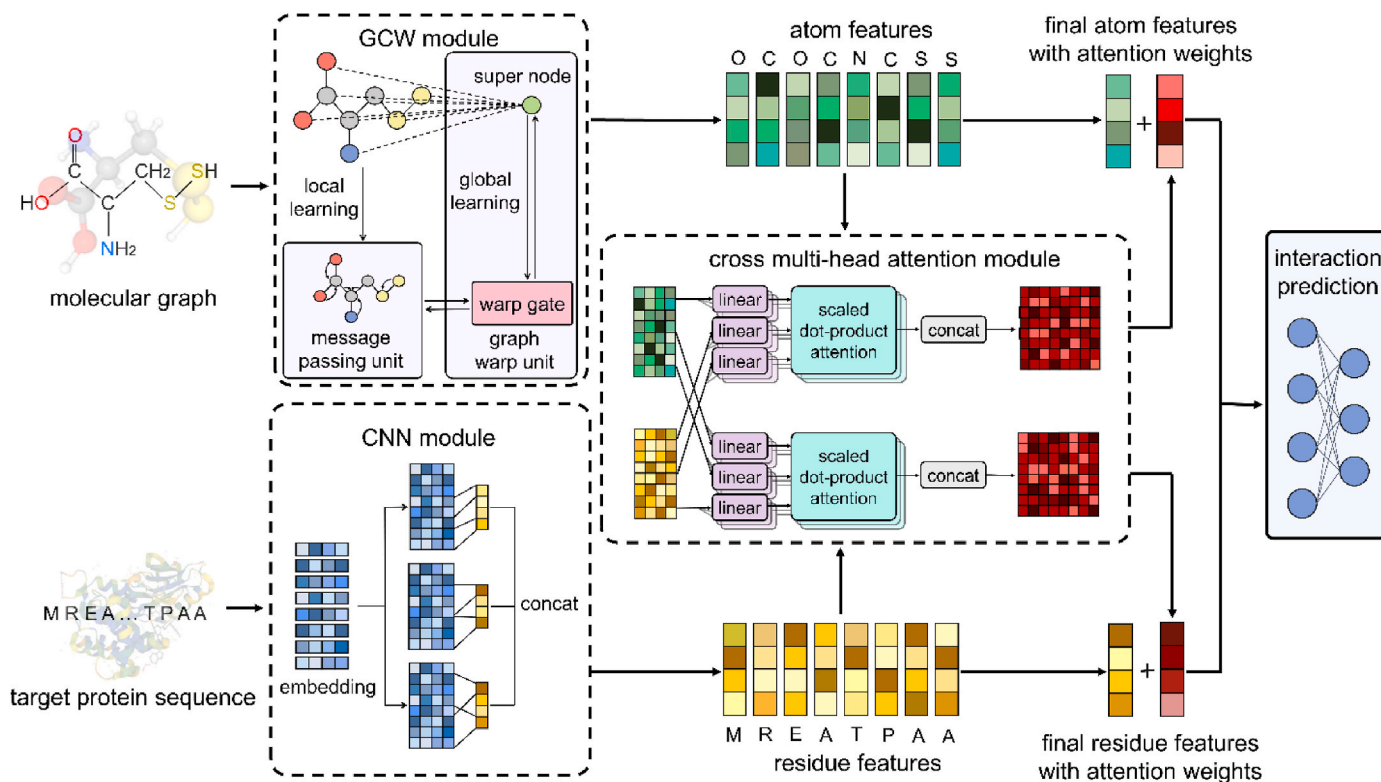


Fig. 1. The schematic workflow of CmhAttCPI. CmhAttCPI takes molecular graphs and protein sequences as inputs. The molecular graph is fed into the GCW module to learn atom features, and the sequence is fed into the CNN module to learn residue features. Then, the extracted atom and residue features are sent to the cross multi-head attention module to generate atom and residue attention weights. The final atom and residue features with attention weights are inputted into FCNN to predict compound–protein interaction.

features as input, a FCNN is applied to predict probabilities of compound-protein interactions.

We conducted a series of 5-fold cross validation experiments on four public CPI datasets, which include both balanced and imbalanced datasets, to evaluate the performance of CmhAttCPI. The experimental results signify that CmhAttCPI consistently outperforms other state-of-the-art models. To comprehensively evaluate the generalization ability of CmhAttCPI, we constructed three evaluation scenarios based on compound and protein clustering. The evaluation results in three evaluation scenarios affirm the outstanding generalization ability of CmhAttCPI. In addition, we visualized atom and residue attention weights and associated the residues with high attention weights to specific biological functions, providing explanations for compound-protein interactions from both the chemical and biological perspective. In case studies, we applied CmhAttCPI to predict compounds for anti-breast cancer and identify anti-cancer candidate compounds. The findings indicate that CmhAttCPI has the strong practical applicability in screening anticancer candidate compounds. The main contributions of this study are summarized as follows.

- (i) We introduce CmhAttCPI, a cross multi-head attention oriented CPI prediction model that can provide chemical and biological interpretation for compound-protein interactions.
- (ii) We comprehensively evaluated the generalization ability of CmhAttCPI in three evaluation scenarios constructed based on compound and protein clustering.
- (iii) In case studies, we employed CmhAttCPI to predict anti-cancer compounds, demonstrating the practical applicability of the proposed model.

2. Materials and methods

2.1. Materials

In this study, we compared the CPI prediction performance of CmhAttCPI with other state-of-the-art deep learning frameworks on four public CPI datasets: BindingDB, DrugBank, GPCR, and Davis. In addition, we conducted further experiments on BindingDB dataset to comprehensively evaluate the generalization performance of CmhAttCPI in different unknown settings. BindingDB contains 33,772 positive CPIs and 27,486 negative CPIs between 49,745 compounds and 812 proteins [24]. The DrugBank dataset was curated from the DrugBank database (version 5.1.10) [25], containing 15,153 known positive compound-protein pairs between 5,162 compounds and 2,668 proteins. To obtain a balanced dataset, we followed a common practice of randomly sampling from unknown pairs as negative CPIs, ensuring an equal number with the positive pairs [26]. The GPCR dataset was constructed from the GLASS database, where all proteins are G protein-coupled receptor (GPCR) family [27]. In GPCR dataset, samples with binding affinities (e.g., K_i , K_d , IC_{50} and EC_{50}) below $10 \mu M$ are considered as positive samples and higher than $10 \mu M$ are negative samples. The duplicate data overlapping with BindingDB in GPCR dataset were removed. The Davis dataset records selectivity assay K_d values measuring binding affinities among kinase proteins and relevant compounds. Divided into positive samples and negative samples by K_d threshold of $10 \mu M$, the Davis dataset contains 30,056 interactions of 68 compounds and 442 kinase proteins. See Table 1 for a detailed data

Table 1
Summary of four public CPI datasets.

Dataset	Proteins	Compounds	Interactions	Positive	Negative
BindingDB	812	49,745	61,258	33,772	27,486
DrugBank	2,668	5,162	30,306	15,153	15,153
GPCR	556	134,230	305,432	256,380	49,052
Davis	68	442	30,056	9,125	20,931

description.

2.2. Methods

CmhAttCPI consists of four modules: (i) graph convolutional networks with a graph warp unit (GCW) module for molecular representations; (ii) convolutional neural network (CNN) module for protein representations; (iii) cross multi-head attention module for capturing the important atoms and residues; (iv) a FCNN for CPI identification.

2.2.1. Molecular representations

Graph convolutional networks (GCNs) is a powerful model in the analysis of molecular graphs in medicinal science [28,29]. Recently, a large number of GCN models have been proposed for learning the molecular features from graph representations of compounds [30,31]. In GCW module, we attached a graph warp unit to the generic GCNs, which can improve its representing power [32]. As depicted in Fig. 2, the GCW module consists of a message passing unit and a graph warp unit. This module not only captures local features from neighboring nodes through the message passing unit but also acquires a global feature of the graph representation via the graph warp unit. To efficiently extract the global feature of the input compound, we introduced a super node in the graph warp unit to connect with all nodes in the graph representation. The extracted global feature is used to updated atom features, promoting the representing power of the GCW module.

We first defined the graph representations of compounds. An input compound with N atoms can be represented by a graph $G = (V, E)$, where $v_i \in V$, $i = 1, 2, \dots, N$, is the i -th atom in the molecule, and $e_{ij} \in E$, $i, j \in \{1, 2, \dots, N\}$, is the bond between the i -th and j -th atom. More specifically, each node $v_i \in V$ is initially represented by a feature vector v_i^{init} of length 82, which is the concatenation of one-hot encodings representing the atom type, degree, explicit valence, implicit valence and whether the atom is in an aromatic structure. Each edge $e_{ij} \in E$ is represented by a feature vector e_{ij} of length 6, which is the concatenation of one-hot encodings representing the bond type and other properties.

Then, the updated atom features v_i^l are produced by L iterations of the GCW module processing the atom features and super node feature, which is the global feature for the input graph representation. The atom features are unceasingly updated utilizing a message passing unit and a graph warp unit at each iteration of the GCW module. The message passing unit extracts local features from the graph representation by executing gathering information and updating information. In the gathering information of l -th iteration, each atom v_i gathers local information from its neighboring atoms and bonds, the information propagating is as follows:

$$h_i^l = \sum_{v_j \in N^{(i)}} f(W_g^l < v_j^{l-1}, e_{ij} >) \quad (1)$$

where $l \in \{1, 2, \dots, L\}$, $W_g^l \in \mathbb{R}^{d_h \times (d_h+6)}$ is the gathering weight matrix. d_h is the hidden size. h_i^l represents the local information of atom v_i in the l -th iteration, $N^{(i)}$ represents the set of neighboring atoms of v_i , v_j^{l-1} represents the feature of atom v_j from the $(l-1)$ -th iteration, e_{ij} is the bond between the i -th atom v_i and j -th atom v_j . $f(\cdot)$ is the ReLU activation function and $< \cdot, \cdot >$ stands for the concatenation. In the updating information of l -th iteration, each atom feature is updated using the gathered information and the atom features learned from previous iterations. The process of updating information is as follows:

$$o_i^l = f(W_u^l < h_i^l, v_i^{l-1} >) \quad (2)$$

where $W_u^l \in \mathbb{R}^{d_h \times 2d_h}$ is the updating weight matrix. o_i^l represents the updated local feature of v_i in the l -th iteration.

The graph warp unit captures global feature of the graph representation by constructing a virtual super node g^0 , which is responsible for

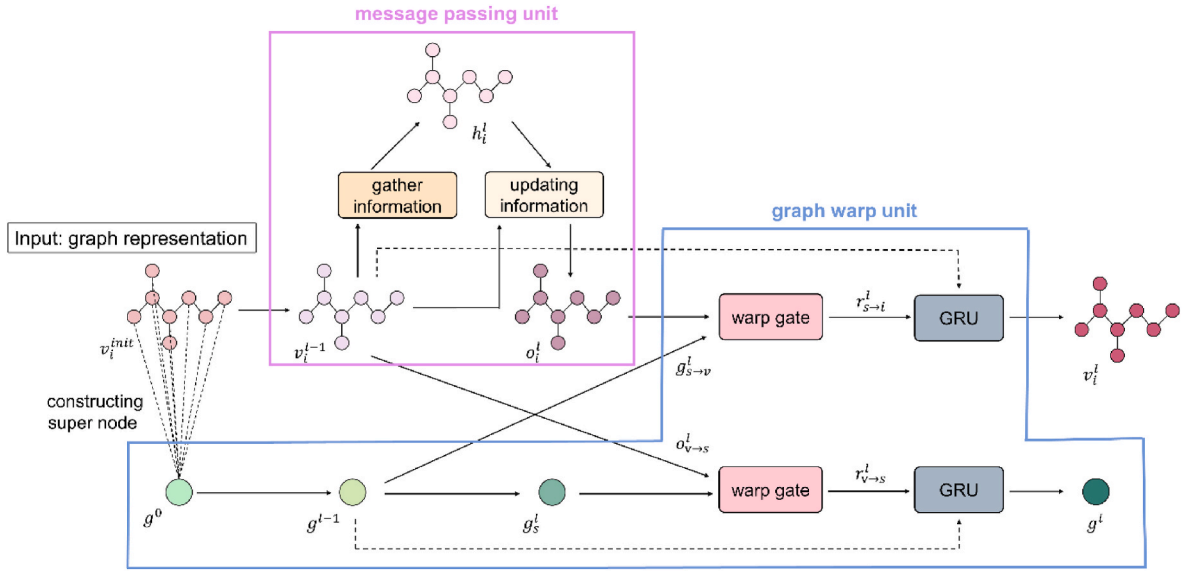


Fig. 2. Diagram of the GCW module.

promoting the remote message passing by communicating with all nodes in the graph. The super node feature g^0 needs to be initialized by summing up all atom features before L iterations:

$$g^0 = \sum_{i=1}^N v_i^0 \quad (3)$$

where v_i^0 is the transformed initial atom feature of v_i , which is calculated by a single-layer neural network:

$$v_i^0 = f(W_{init} v_i^{init}) \quad (4)$$

where v_i^{init} stands for the initial atom feature of v_i , which is 82-dimensional one-hot feature vector. $f(\cdot)$ is the ReLU activation function, and $W_{init} \in \mathbb{R}^{h_d \times 82}$ is weight matrix of the single-layer neural network.

Distant atoms in the graph can communicate effectively by transmitting messages between all the atoms in the graph and the super node, and such a global feature can be captured. First, the super node feature is updated in the graph warp unit applying a single-layer neural network to get g_s^l :

$$g_s^l = \tanh(W_{super}^l g^{l-1}) \quad (5)$$

where, g^{l-1} is the super node feature in the $(l-1)$ -th iteration, W_{super}^l is the updating weight matrix and $\tanh(\cdot)$ is the tanh activation function.

Then, the final updated atom features and super node feature at each iteration of GCW module are obtained by the following three steps: (1) gathering information from the super node to all of atoms and gathering information from all of atoms to the super node; (2) passing information using warp gates; (3) updating atom features and the super node feature by gated recurrent units (GRUs). In step (1), the gathering information $g_{s \rightarrow v}^l$ from the super node to atoms is simply given by:

$$g_{s \rightarrow v}^l = \tanh(W_{s \rightarrow v}^l g^{l-1}) \quad (6)$$

where $W_{s \rightarrow v}^l \in \mathbb{R}^{h_d \times h_d}$ is weight matrix and g^{l-1} is the super node feature in the $(l-1)$ -th iteration. In the gathering information from atoms to the super node, a K-head attention mechanism is applied to calculate contribution weights of individual atoms. The information $o_{v \rightarrow s}^l$ gathered from each atom to the super node is calculated:

$$o_{v \rightarrow s}^l = \tanh\left(W_{v \rightarrow s}^l < \sum_{i=1}^N \alpha_i^{l,1} v_i^{l-1}, \sum_{i=1}^N \alpha_i^{l,2} v_i^{l-1}, \dots, \sum_{i=1}^N \alpha_i^{l,K} v_i^{l-1} >\right) \quad (7)$$

$i \in \{1, 2, \dots, N\}$, $l \in \{1, 2, \dots, L\}$, $k \in \{1, 2, \dots, K\}$, N, L, K are the number of atoms, iterations and heads, respectively. $\alpha_i^{l,k}$ is an attention weight attaching the i -th atom v_i at the k -th head in the l -th iteration, $< \cdot, \cdot >$ stands for the concatenation.

In step (2), the information passed to each atom can be obtained through the combination of the updated atom features o_i^l in the message passing unit and the gathering information $g_{s \rightarrow v}^l$ from the super node to atoms by the warp gate $z_{s \rightarrow i}^l$ in the l -th iteration:

$$r_{s \rightarrow i}^l = (1 - z_{s \rightarrow i}^l) \odot o_i^l + z_{s \rightarrow i}^l \odot g_{s \rightarrow v}^l \quad (8)$$

$$z_{s \rightarrow i}^l = \sigma(W_H^l o_i^l + W_G^l g_{s \rightarrow v}^l) \quad (9)$$

where, $r_{s \rightarrow i}^l$ is the information passed to each atom, $l \in \{1, 2, \dots, L\}$, $W_H^l, W_G^l \in \mathbb{R}^{h_d \times h_d}$ are weight matrices, $\sigma(\cdot)$ and $\odot$ stand for sigmoid activation function and element-wise product, respectively. In the same vein, the information passed to super node can be produced through the combination of the updated super node feature g_s^l in the graph warp unit and the gathering information $o_{v \rightarrow s}^l$ from each atom to the super node by the warp gate $z_{v \rightarrow s}^l$ in the l -th iteration:

$$r_{v \rightarrow s}^l = (1 - z_{v \rightarrow s}^l) \odot o_{v \rightarrow s}^l + z_{v \rightarrow s}^l \odot g_s^l \quad (10)$$

$$z_{v \rightarrow s}^l = \sigma(W_H^{l(S)} o_{v \rightarrow s}^l + W_G^{l(S)} g_s^l) \quad (11)$$

where, $r_{v \rightarrow s}^l$ is the information passed to the super node, $W_H^{l(S)}, W_G^{l(S)} \in \mathbb{R}^{h_d \times h_d}$ are weight matrices. In step (3), atom features v_i^l and super node features g^l are updated by using two GRUs:

$$v_i^l = GRU(v_i^{l-1}, r_{s \rightarrow i}^l), g^l = GRU(g^{l-1}, r_{v \rightarrow s}^l) \quad (12)$$

where v_i^{l-1} and g^{l-1} stand for the feature of atom v_i and the super node feature in the $(l-1)$ -th iteration, $r_{s \rightarrow i}^l$ and $r_{v \rightarrow s}^l$ stand for the information passed to atom v_i and the information passed to the super node, respectively.

2.2.2. Protein representations

We used convolutional neural networks (CNNs) for protein sequences. First, we constructed “vocabulary” for protein sequences, each residue is encoded with an integer. In this manner, a protein sequence can be represented as an integer sequence. Then the integer sequence is transformed into an embedding matrix by an embedding layer.

CNNs are well-known deep learning architectures that contain one or more convolutional layers. In the last few years, CNNs have achieved significant results on medical tasks swiftly [33,34]. CNNs can capture the powerful local dependencies by kernels. Therefore, the types of features are directly affected by the sizes of the kernels in CNNs. In CmhAttCPI, the CNN module is constructed based on multiple parallel convolutional layers with different kernels, facilitating the learning of more protein sequence features from different views. The output hidden vector of convolutional layer is represented as follows:

$$c_k = f(\sigma_{bn}(P_{embed}W_k + b_k)) \quad (13)$$

where $P_{embed} \in \mathbb{R}^{l_p \times d_e}$ is embedding matrix of the given protein, l_p is sequence length of the protein and d_e is embedding hidden size. $W_k \in \mathbb{R}^{d_e \times d}$ and b_k are weight matrix and bias in the convolutional layer, respectively. $k \in \{1, 2, \dots, K\}$ is the number of different convolution kernels. σ_{bn} stands for BatchNorm operation. $f(\cdot)$ is the *Elu* activation function. Finally, the representation of the protein is obtained by concatenate the hidden vectors:

$$p = \langle c_1, c_2, \dots, c_K \rangle \quad (14)$$

2.2.3. Cross multi-head attention module

Attention mechanism is one of the most important concepts in the deep learning field and was first introduced into natural language processing (NLP) through machine translation tasks [16,35]. They can provide certain interpretability about the contributions of individual input positions to the final prediction results by calculating attention weight for each input position [36]. We designed a special attention module, called cross multi-head attention, revealing the important atoms and residues contributing significantly to compound-protein interactions. The cross multi-head attention module integrates two multi-head scaled dot-product attentions, P_to_C attention and C_to_P attention, to learn the contributions of atoms and residues to the interactions, respectively. P_to_C attention is used to measure the contributions of different residues to each atom, and C_to_P attention is applied to measure the contributions of different atoms to each residue.

A multi-head scaled dot-product attention consists of multiple par-

allel scaled dot-product attention layers. Fig. 3 depicts the details of a scaled dot-product attention layer. A scaled dot-product attention layer can transform a query and a set of key-value pairs into an output that is a weighted sum of the values. The weight assigned to each value is computed by a compatibility function of the query with the corresponding key [16]. The scaled dot-product attention layer takes query (Q), key (K) and value (V) matrix as input. And the input matrix Q , K of dimension d_k and V of dimension d_v are obtained by conducting linear projection on the feature matrix X :

$$Q = XW_Q, K = XW_K, V = XW_V \quad (15)$$

where $X \in \mathbb{R}^{l \times d}$, assuming that X represents a sentence, l stands for numbers of words in the sentence and d stands for embedding dimension. $W_Q \in \mathbb{R}^{d \times d_k}$, $W_K \in \mathbb{R}^{d \times d_k}$, $W_V \in \mathbb{R}^{d \times d_v}$ are weight matrices. We calculated the dot products of the query with all keys and divide each by $\sqrt{d_k}$. Subsequently, a softmax function is applied to obtain the weights on the values:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (16)$$

As displayed in Fig. 4, the cross multi-head attention module incorporates P_to_C attention and C_to_P attention, strategically designed to encapsulate richer encoded representation information in different subspaces significantly enhancing the expressive capacity of the model. P_to_C attention is applied to estimate the contributions of different residues of the protein to each atom of the compound. In P_to_C attention, X_C is used to produce query matrix and X_P is used to produce key and value matrix. First, X_C is projected H times to obtain Q_1^h and X_P is projected H times to K_1^h and V_1^h :

$$Q_1^h = X_C W_{Q_1}^h, K_1^h = X_P W_{K_1}^h, V_1^h = X_P W_{V_1}^h \quad (17)$$

where $h \in \{1, 2, \dots, H\}$, H is the number of parallel scaled dot-product attention layers or heads, $X_C \in \mathbb{R}^{l_C \times d_C}$ and $X_P \in \mathbb{R}^{l_P \times d_P}$ represent the given compound and protein feature matrix produced from GCW and CNN module, l_C and l_P stand for numbers of atoms in the input compound and residues in the input protein, d_C and d_P represent the feature dimensions of the compound and protein, respectively. $W_{Q_1}^h \in \mathbb{R}^{d_C \times d_{Q_1}^h}$, $W_{K_1}^h \in \mathbb{R}^{d_P \times d_{K_1}^h}$, $W_{V_1}^h \in \mathbb{R}^{d_P \times d_{V_1}^h}$ are weight matrices, $d_{Q_1}^h = \frac{d_C}{H}$ and $d_{K_1}^h = d_{V_1}^h = \frac{d_P}{H}$. Then, we performed the attention function of scaled dot-product attention layer defined by Equation (15) on each of these projections Q_1^h, K_1^h, V_1^h :

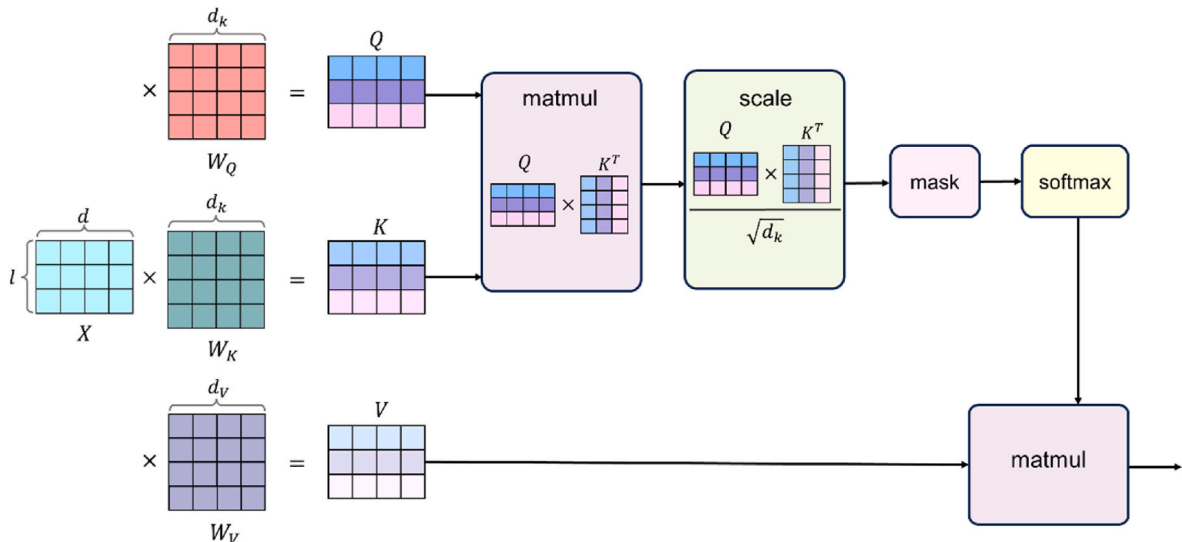


Fig. 3. Diagram of a scaled dot-product attention layer.

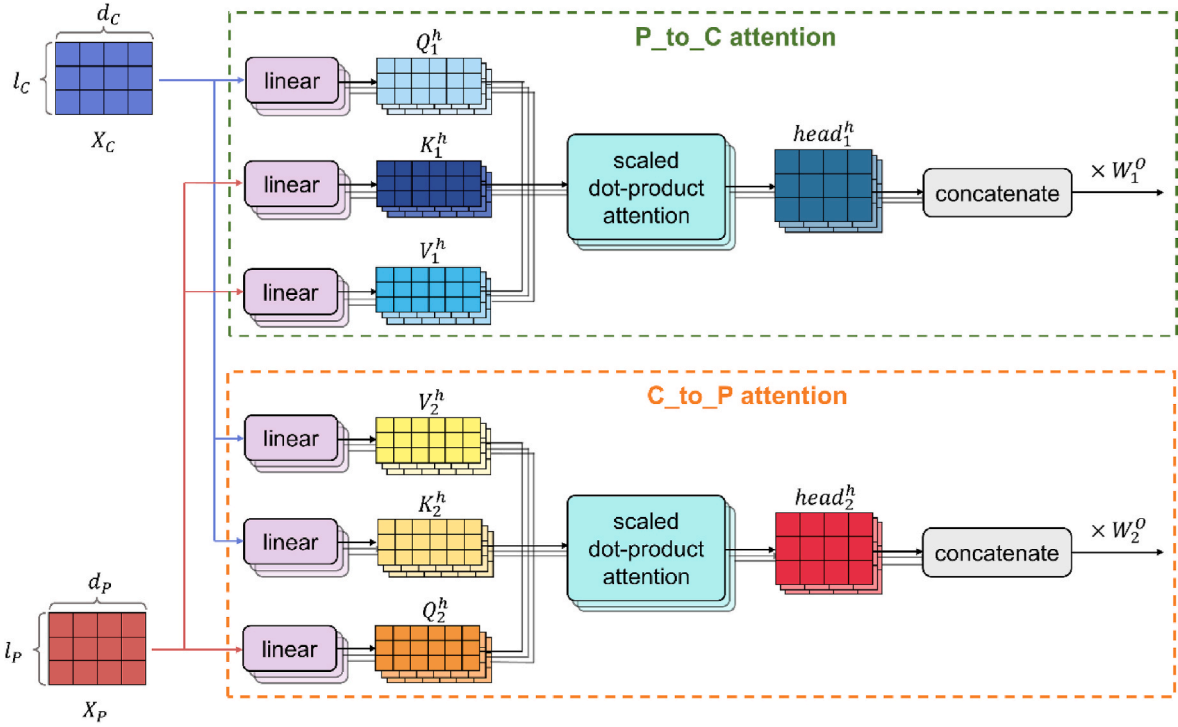


Fig. 4. Diagram of the cross multi-head attention module.

$$head_1^h = \text{Attention}(Q_1^h, K_1^h, V_1^h) \quad (18)$$

Finally, the output feature of P_to_C attention is calculated by concatenating these output of attention functions and performing a linear transformation:

$$\text{multihead}(Q_1^{1,\dots,H}, K_1^{1,\dots,H}, V_1^{1,\dots,H}) = \langle head_1^1, \dots, head_1^H \rangle W_1^O \quad (19)$$

where $h \in \{1, 2, \dots, H\}$, H is the number of heads and $\langle \cdot, \cdot \rangle$ stands for the concatenation. $W_1^O \in \mathbb{R}^{Hd_h \times d_p}$ is the dot product weight and $d_v = \frac{d_p}{H}$.

Similarly, C_to_P attention is applied to estimate the contributions of different atoms of the compound to each residue of the protein. In C_to_P attention, X_P is used to produce query matrix and X_C is used to produce key and value matrix. First, X_P is projected H times to obtain Q_2^h and X_C is projected H times to K_2^h and V_2^h :

$$Q_2^h = X_P W_{Q_2}^h, K_2^h = X_C W_{K_2}^h, V_2^h = X_C W_{V_2}^h \quad (20)$$

where $W_{Q_2}^h \in \mathbb{R}^{d_p \times d_q}$, $W_{K_2}^h \in \mathbb{R}^{d_c \times d_k}$, $W_{V_2}^h \in \mathbb{R}^{d_c \times d_v}$ are weight matrices, $d_q = \frac{d_p}{H}$ and $d_k = d_v = \frac{d_c}{H}$. The attention function is conducted on each of these projections Q_2^h, K_2^h, V_2^h :

$$head_2^h = \text{Attention}(Q_2^h, K_2^h, V_2^h) \quad (21)$$

The output feature of C_to_P attention is obtained as follows:

$$\text{multihead}(Q_2^{1,\dots,H}, K_2^{1,\dots,H}, V_2^{1,\dots,H}) = \langle head_2^1, \dots, head_2^H \rangle W_2^O \quad (22)$$

where $h \in \{1, 2, \dots, H\}$, H is the number of heads and $\langle \cdot, \cdot \rangle$ stands for the concatenation. $W_2^O \in \mathbb{R}^{Hd_v \times d_c}$ is the dot product weight and $d_v = \frac{d_c}{H}$.

2.2.4. Fully connected neural network for CPI prediction

The final atom and residue features are obtained by adding the weights of P_to_C attention to the atom features from the GCW module and adding the weights of C_to_P attention to the residue features from the CNN module, respectively. At last, linear projections in FCNN are used to further calculate the probability $\hat{y}$ of the input compound-protein interaction:

$$\hat{y} = f(\langle X_C + M_{att}, X_P + P_{att} \rangle W_{output} + b_{output}) \quad (23)$$

where $X_C \in \mathbb{R}^{l_C \times d_C}$ and $X_P \in \mathbb{R}^{l_P \times d_P}$ stand for the compound and protein feature matrix from GCW and CNN module, respectively. $M_{att} \in \mathbb{R}^{l_C \times d_C}$ and $P_{att} \in \mathbb{R}^{l_P \times d_P}$ represent the output attention weight matrix from P_to_C attention and C_to_P attention, respectively. $f(\cdot)$ is the ReLU activation function and $\langle \cdot, \cdot \rangle$ is the concatenation. $W_{output} \in \mathbb{R}^{(d_C + d_P) \times d}$ and b_{output} are the weight matrix and the bias vector, respectively. d is the number of units in the hidden layer. As CPI prediction is a binary classification task, we used binary cross entropy loss to train CmhAttCPI:

$$loss = -[y \log \hat{y} + (1 - y) \log (1 - \hat{y})] \quad (24)$$

3. Experiments

3.1. Scenarios based on clustering

To comprehensively evaluate the generalization performance of CmhAttCPI, we constructed three evaluation scenarios based on compound and protein clustering, including compound clustering unknown setting, protein clustering unknown setting and both unknown setting. Each scenario has multiple conditions by setting different clustering thresholds. The experiments in all settings are conducted using cluster-cross validation [23], an advancement of cross validation that distributes entire clusters of compounds or proteins across folds instead of randomly assigning data points to folds as in 5-fold cross validation.

First, we clustered compounds and proteins using single-linkage clustering algorithm [37], more details are illustrated in Section 1.1 of the supplementary material. We used clustering thresholds of 0.3, 0.4, and 0.5. A small threshold cannot achieve perfect separation to avoid data redundancy caused by similarities of compounds and proteins, and a large threshold leads to excessively large clusters resulting in a greatly imbalanced separation between the training and test sets, that is, there are too large (small) samples in the training set and too small (large) in the test set. The size of compound and protein clusters under different thresholds are displayed in Supplemental Tables S1 and S2.

Then, the cluster-cross validation is conducted in three evaluation scenarios on compound and protein clusters. In cluster-cross validation, the splitting ratio of the training-validation and test sets is 4:1. Note that the ratio is an approximation, because the splitting is performed on compound and protein clusters, and the number of samples among individual clusters is not necessarily evenly distributed. The validation set is generated by randomly selecting 12.5% samples of the training-validation set. Such a splitting strategy results in an approximate ratio of 7:1:2 for the training set, validation set and test set.

In the compound clustering unknown setting, the splitting of the training-validation and test sets ensures that the compounds within the same cluster are either all used in the training-validation set, or all used in the test set. Similarly, in the protein clustering unknown setting, the splitting of the training-validation and test sets ensures that the proteins within the same cluster are either all included in the training-validation set, or all included in the test set. In the both unknown setting, both the compounds from the same cluster and proteins from the same cluster are either all present in training-validation set or test set. And more details of evaluation scenarios are illustrated in Section 1.2 of the supplementary material.

3.2. Training and deployment

Training and deployment of CmhAttCPI was performed using a workstation equipped with two NVIDIA Quadro GV100 32 GB GPU cards. Our model was implemented with Pytorch 1.7.1. CmhAttCPI takes molecular graphs as input, which are converted from SMILES strings by RDKit [38]. For the training, we used the Adam optimizer. The learning rate and the batch size were set to 0.0001 and 128, respectively.

3.3. Evaluation metrics

In this work, we applied the area under the receiver operating characteristics curve (AUC) and accuracy to measure the performance of the proposed model CmhAttCPI and baseline models. The value of AUC is close to 1, which indicates the performance of model is perfect,

whereas for simple random predictive, AUC is close to 0.5. Accuracy is calculated based on a confusion matrix as follows:

$$accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (25)$$

where TP is true positive, TN is true negative, FP is false positive, FN is false negative.

4. Results

4.1. Performance evaluation

4.1.1. Performance evaluation on balanced datasets

Drug repurposing is the widest application for CPI prediction. It is beneficial if a drug that has successfully approved by the FDA could treat other new diseases in view of development cost and safety [39]. Drug repurposing is built based the assumption that drugs often interact with multiple protein targets [40]. Considering this situation, we evaluated the performance of CmhAttCPI against other six baseline methods for CPI prediction on two balanced datasets (BindingDB, DrugBank) and two imbalanced datasets (GPCR, Davis) to test effectiveness of the model. The summary and hyperparameter settings of the baseline models are displayed in Section 1.3 of the supplementary material. The descriptions of these public datasets are shown in Table 1.

The experiments on the balanced datasets are conducted using 5-fold cross-validation, and the results are presented in Fig. 5. CmhAttCPI consistently outperforms the baselines in terms of average AUC of 0.9778 on BindingDB dataset, with standard deviation (SD) of 0.0008. Similarly, excellent performance is also observed on DrugBank dataset, where the average AUC of CmhAttCPI is 0.8507, with SD of 0.0047. The ROC and PR curves of CmhAttCPI and other models on balanced datasets can be found in Supplemental Fig. S1.

4.1.2. Performance evaluation on imbalanced datasets

We further evaluated the effectiveness of CmhAttCPI on two imbalanced datasets, GPCR and Davis. The GPCR dataset exhibits a higher

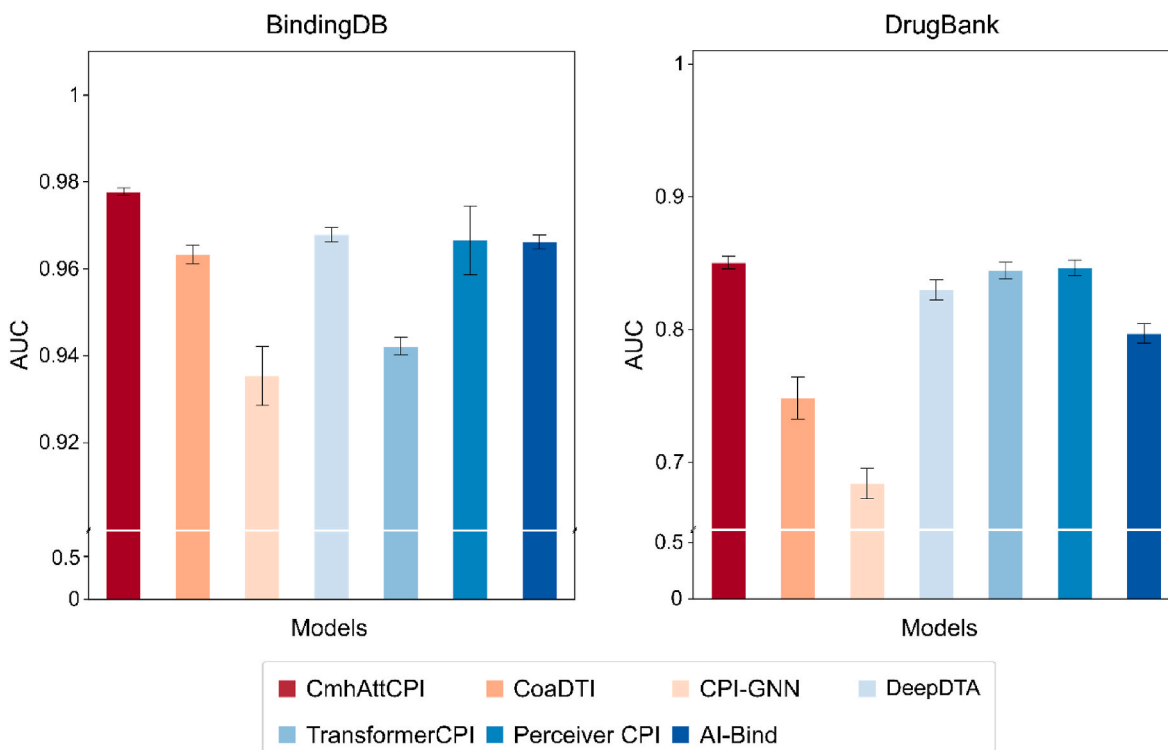


Fig. 5. Performance evaluation of CmhAttCPI and baseline methods on balanced datasets.

proportion of positive samples compared to negative samples, with a positive-to-negative interaction ratio of 5.2:1 (Table 1). In contrast, the Davis dataset shows a higher proportion of negative samples compared to positive samples, with a positive-to-negative interaction ratio of 1:2.3 (Table 1). The evaluation results on imbalanced datasets are depicted in Fig. 6, we see that the proposed model outperforms other baselines on imbalanced datasets. CmhAttCPI yields an average AUROC of 0.8687 and AUPR of 0.9660 on GPCR, with SD of 0.0031 and 0.0007, respectively (Fig. 6A). Similar outstanding performance is presented on Davis dataset, with average AUROC and AUPR of 0.8688 and 0.7807, and SD of 0.0100 and 0.0148, respectively (Fig. 6B). The above experimental results indicate that the proposed model not only achieves excellent performance on balanced datasets but also consistently maintains high effectiveness and robustness on imbalanced datasets.

4.2. Generalization evaluation

In real datasets for CPI prediction, there often exist highly similar compounds and proteins. The similarities between compounds and proteins can frequently result in the overestimation of the model generalization performance. To comprehensively evaluate the generalization ability of our framework, we conducted cluster-cross validation [23] in three evaluation scenarios: compound clustering unknown setting, protein clustering unknown setting and both unknown setting.

Compound clustering unknown setting has practical significance when identifying potential unknown targets that may interact with newly discovered chemical molecules, especially in cases where the 3D structures of target proteins and high-quality negative samples are

unavailable. Protein clustering unknown setting is conducive to identifying new therapeutic targets for complicated diseases, elucidating the mechanism of action of compounds and discovering other unknown side effects [41,42]. Both unknown setting also plays a crucial role in realistic drug discovery.

The comparison results of AUC and accuracy in three scenarios with different clustering thresholds are depicted in Fig. 7. The ROC and PR curves of CmhAttCPI and other models in three scenarios with different clustering thresholds are displayed in Supplemental Figs. S2–S4.

In compound clustering unknown setting, CmhAttCPI consistently and significantly outperforms other models in terms of the average AUC and accuracy with different clustering thresholds (Fig. 7A). In protein cluster unknown setting, CmhAttCPI presents the highest AUC and accuracy with the thresholds of 0.3 and 0.4 (Fig. 7B). Although AI-Bind (0.7173 ± 0.0195) exhibits better average accuracy than CmhAttCPI (0.6901 ± 0.0302) with the threshold of 0.5, CmhAttCPI achieved a significantly higher AUC (0.7894 ± 0.0194) compared to AI-Bind (0.741 ± 0.0373). In both unknown settings, CmhAttCPI also shows superior performance to other models with different thresholds (Fig. 7C). These results prove the generalization ability of CmhAttCPI to infer potential targets interacting with novel molecules and discover new therapeutic targets and side effects for known drugs.

4.3. Ablation study

We conducted ablation experiments by removing relevant sub-structures from CmhAttCPI to confirm their contribution to performance improvement. These experiments are performed based on BindingDB

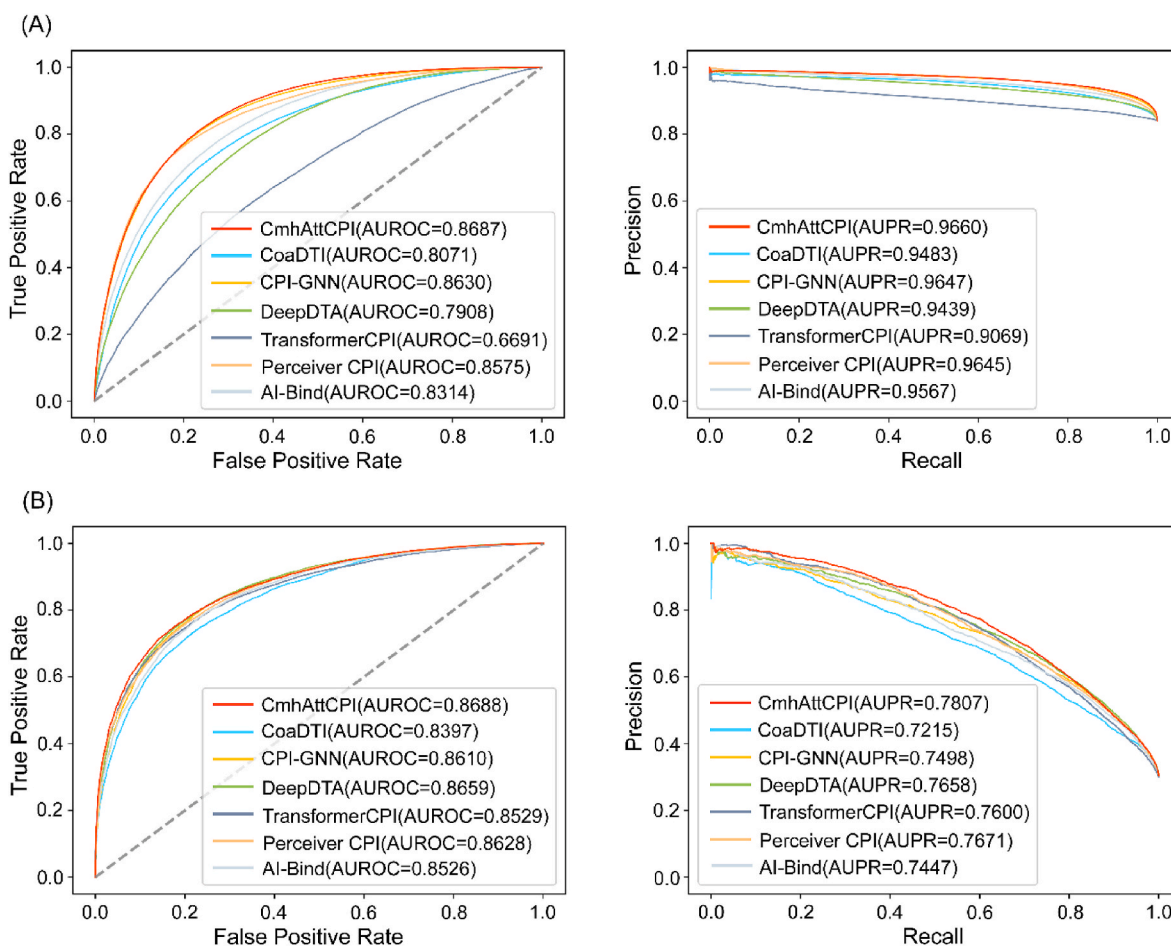


Fig. 6. Performance evaluation of CmhAttCPI and baseline methods on imbalanced datasets. (A) ROC and PR curves on GPCR dataset; (B) ROC and PR curves on Davis dataset.

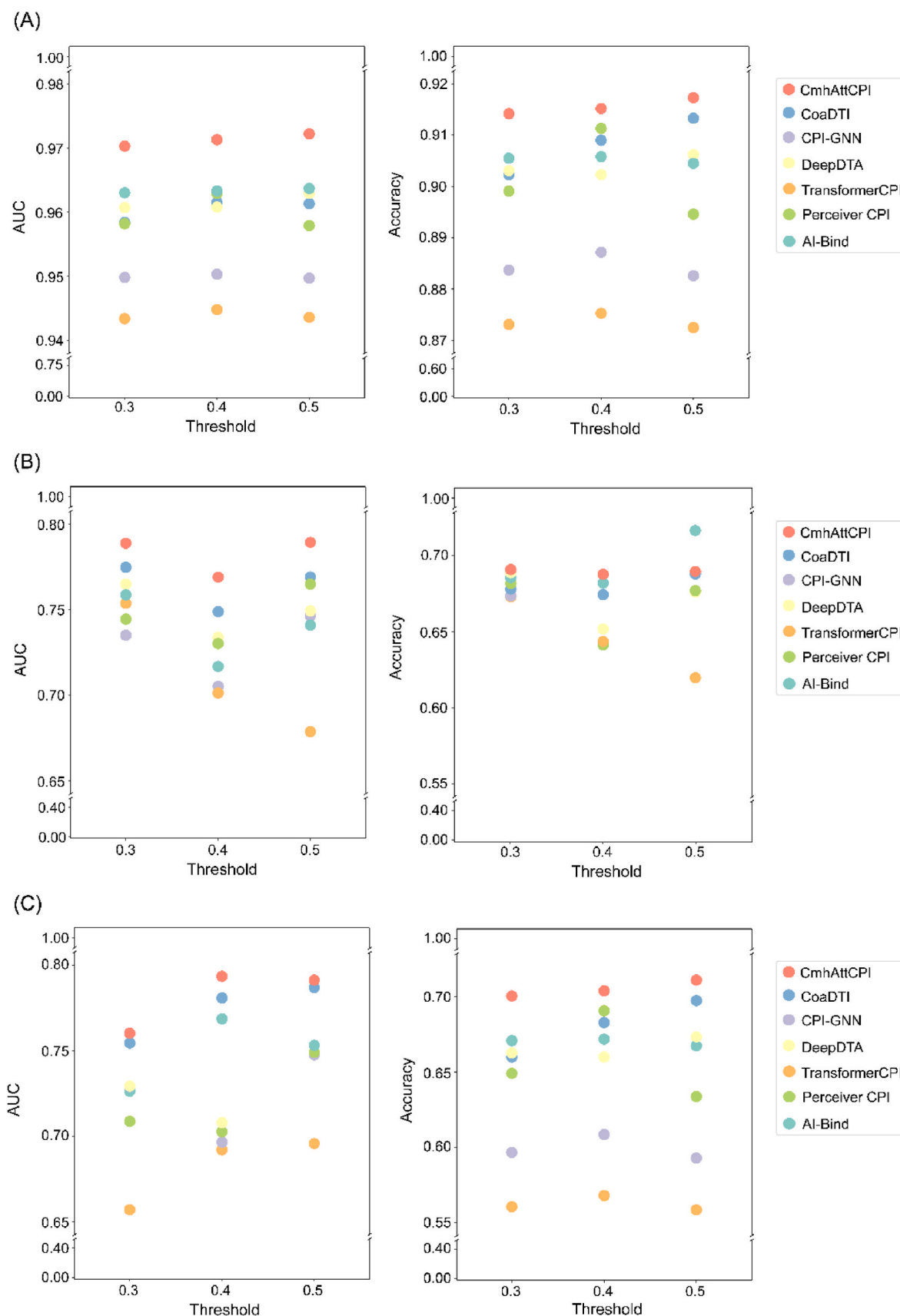


Fig. 7. Generalization performance evaluation in three evaluation scenarios with different clustering thresholds. (A) Performance evaluation in compound clustering unknown setting; (B) performance evaluation in protein clustering unknown setting; (C) performance evaluation in both unknown setting.

dataset with the following configurations: (i) -Warp indicates the removal of the graph warp unit from the GCW module (Supplemental Fig. S5); (ii) -Cmh_Attention indicates the removal of the cross multi-head attention module (Supplemental Fig. S6); (iii) -P_to_C_Attention indicates the removal of P_to_C attention from the cross multi-head attention module (Supplemental Fig. S7). (iv) -C_to_P_Attention indicates the removal of C_to_P attention from the cross multi-head attention module (Supplemental Fig. S8).

Relying on the results presented in Table 2, we observe that the graph warp unit, cross multi-head attention module, P_to_C attention and C_to_P attention play crucial roles in performance improvement of the model. CmhAttCPI outperforms the -Warp configuration, confirming that the graph warp unit enhances performance by efficiently learning the molecular global features. Moreover, CmhAttCPI performs better than the -Cmh_Attention, -P_to_C_Attention and -C_to_P_Attention configurations, indicating the significance of the cross multi-head attention module consisting of two multi-head attentions in enhancing predictive performance.

4.4. Model interpretation

We selected Bcl-2 and p38 MAP kinase as target proteins. Bcl-2, a member of protein family involved in regulating cell apoptosis, necrosis, and autophagy [43], is implicated in many human diseases, particularly cancer [44]. The p38 MAP kinase is of vital importance in regulating the production of proinflammatory cytokines, such as tumor necrosis factor [45]. We analyzed Navitoclax-Bcl2 interaction (PDB ID: 4LVT) [46] and BIRB 796-p38 interaction (PDB ID: 1KV2) [47] to demonstrate the bidirectional interpretability of CmhAttCPI in predicting CPIs from both the chemical and biological perspective.

For each example, the compound attention matrix $M_{att} \in \mathbb{R}^{l_c \times d_c}$ and protein attention matrix $P_{att} \in \mathbb{R}^{l_p \times d_p}$ are obtained from P_to_C attention and C_to_P attention in the cross multi-head attention module, respectively. We conducted the mean operator to $M_{att} \in \mathbb{R}^{l_c \times d_c}$ and $P_{att} \in \mathbb{R}^{l_p \times d_p}$ to get the attention vectors $m \in \mathbb{R}^{l_c}$ for the compound and $p \in \mathbb{R}^{l_p}$ for the protein. The vectors m and p represent atom and residue attention weights, respectively. We visualized the atom and residue attention weights to analyze the impact of atoms and residues on compound-protein interactions. The attention weight visualizations are depicted in Fig. 8.

In the interaction between Navitoclax and Bcl-2, Asp100, Phe101, Asp 108, Phe 109, Met 112, Glu133, Leu134, Asn140, Trp141, Gly142, Arg143, Ala146, Phe195, Leu198 are deemed crucial by CmhAttCPI, as exhibited in Fig. 8A. The crucial residues are surrounded the binding pocket. Especially, Asn140 forms a hydrogen bond of 2.67 Å with the nitrogen atom in Navitoclax, and Trp141 forms another hydrogen bond with the oxygen atom. The other residues interact with Navitoclax by other type of interactions such as halogen bond and salt bridge. Surprisingly, we find that the nitrogen and oxygen atom in Navitoclax forming hydrogen bonds with Asn140 and Trp141 are also assigned high attention weights and highlighted in red, as displayed in Fig. 8B. Moreover, CmhAttCPI assigns high attention weights to the atoms in the main carbon backbone essential for the binding pocket, highlighting these atoms in blue.

Then, we associated the high-attention residues with specific biological functions. Phe195, Leu198 are contained within BH1 functional

domain of Bcl-2, Glu133, Leu134, Asn140, Trp141, Gly142, Arg143, Ala146 are contained within BH2 functional domain, Asp100 and Phe101 are contained within the BH3 functional domain. The residues located in the BH1 and BH2 domains are vital for the survival function of anti-apoptotic proteins Bcl-2 and Bcl-xL and for the interactions between these proteins and pro-apoptotic proteins Bax and Bak [48,49]. The interactions hinder the pro-apoptotic proteins from executing their death-promoting functions [50]. And the BH3 domains are also essential for regulation of pro-apoptotic function, preventing cancer cells from evading apoptosis [51]. Thus, the high-attention residues considered by CmhAttCPI possess the key biological functions.

In the interaction between BIRB 796 and p38 MAP kinase, Val 38, Lys 53, Glu71, Leu 72, Leu 108, Met 109, Gly 110, Leu 156, Ala 157, Val 158, Asp168, Phe169 are identified crucial residues by CmhAttCPI, as displayed in Fig. 8C. We observe that Glu71, Gly 109 and Asp168 form hydrogen bonds of <3 Å with BIRB 796. And the other residues interact with BIRB 796 by specific interactions. It is noteworthy that the oxygen atom in BIRB 796 forming hydrogen bond with Asp168 is also assigned high attention weight and highlighted in red, as shown in Fig. 8D. Other high-attention atoms are highlighted in blue.

A previous research reported a new allosteric binding site for a diaryl urea class of highly potent and selective inhibitors against human p38 MAP kinase [47]. The binding site is in the highly conserved motif Asp-Phe-Gly (Asp168-Phe169-Gly170) within the active site of the kinase and the formation of the binding site requires a large conformational change. Solution studies in the research revealed that BIRB 796 has slow binding kinetics, consistent with the requirement for conformational change. The attention weight visualization of BIRB 796-p38 indicates that Asp168 and Phe169 are assigned high attention weights by CmhAttCPI. Specifically, Asp168 forms a hydrogen bond with the oxygen atom in BIRB 796, consistent with the above experimental results.

Overall, CmhAttCPI not only provides bidirectional interpretation, but the key high-attention residues can also be associated with specific biological functions. These visualizations and biological interpretations reveal that CmhAttCPI has a strong ability to discover crucial atoms and residues contributing significantly to the compound-protein interactions and it can assist researchers in improving binding properties and accelerating the development of anticancer drug discovery.

4.5. Case studies

Breast cancer is the most frequently diagnosed cancer in females, accounting for 23% of the total cancer cases and 14% of cancer deaths [52,53]. Thus, there is an urgent need for rapid drug discovery to effectively treat breast cancer.

To verify the practical applicability of CmhAttCPI, we applied a pre-trained CmhAttCPI to identify potential compounds with anti-breast-cancer properties in the DrugBank database. First, we collected 3,459 investigational or experimental drugs from the DrugBank database. Then, the investigational or experimental drugs and the 29 breast cancer driver genes identified by Matthew et al. comprise the test set [54]. The SMILES strings of drugs are obtained from the DrugBank database and protein sequences are scoured from the Uniprot database. Finally, we utilized a pre-trained model to predict potential anti-breast-cancer candidate compounds among the investigational or experimental drugs.

Table 3 presents the top ten predicted candidate compounds for breast cancer therapy. Among these, three compounds have been validated in vitro, and four compounds have exhibited the ability to treat breast cancer or suppress breast cancer metastasis in published studies [55–57]. We further investigated the background of the candidate compounds, and more details are provided in Supplemental Tables S10 and S11.

We next turn to evaluating the capability of CmhAttCPI for screening potential anticancer drugs. In the evaluation experiment, the test set consists of the collected 3,459 compounds without FDA approval from

Table 2
Ablation study on BindingDB.

Configurations	AUC (SD)	Accuracy (SD)
-Warp	0.9627 (0.0023)	0.9012 (0.0035)
-Cmh_Attention	0.9665 (0.0015)	0.9060 (0.0035)
-P_to_C_Attention	0.9540 (0.0103)	0.8869 (0.0149)
-C_to_P_Attention	0.9707 (0.0038)	0.9105 (0.0070)
CmhAttCPI	0.9778 (0.0008)	0.9291 (0.0028)

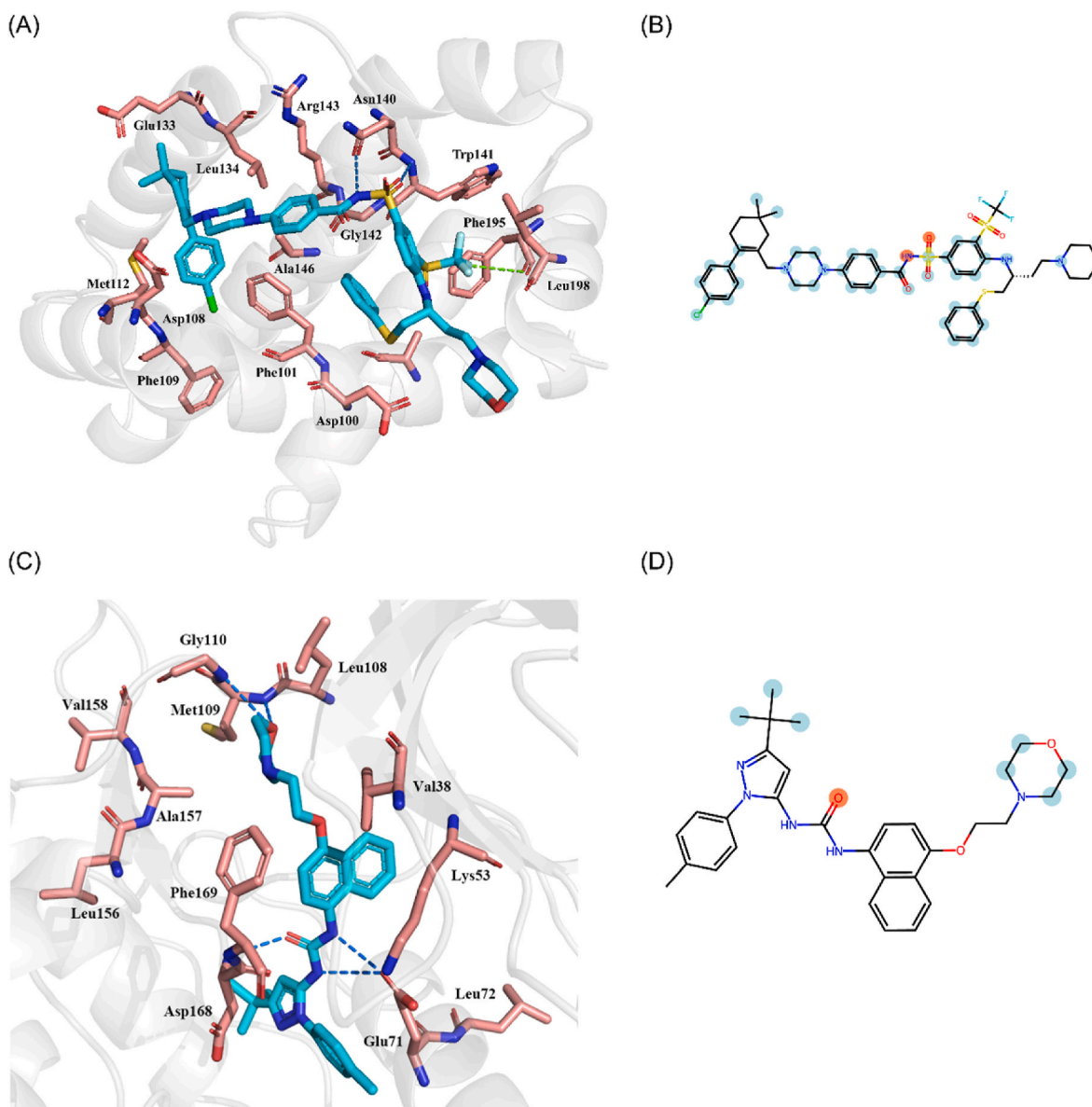


Fig. 8. Attention weight visualizations. (A) The 3D visualization of Navitoclax-Bcl2 binding pocket with high-attention residues; (B) the 2D structure of Navitoclax with high-attention atoms; (C) the 3D visualization of BIRB 796-p38 binding pocket with high-attention residues; (D) the 2D structure of BIRB 796 with high-attention atoms.

the DrugBank database and 235 proteins associated with 32 types of cancers. Subsequently, a pre-trained CmhAttCPI is employed to predict potential anticancer compounds. The top 50 compounds predicted by CmhAttCPI are considered as anticancer candidate compounds, while the last 50 are considered as non-candidate compounds. To explore whether the candidate compounds selected by CmhAttCPI possess anticancer properties, we collected 48 actual anticancer drugs from the DrugBank database and visualized extended connectivity fingerprint (ECFP) [58] features of the candidate compounds, non-candidate compounds and actual anticancer drug with principal component analysis (PCA) [59]. More information of actual anticancer drugs can be found in [Supplemental Table S12](#).

First, the SMILES strings of compounds are converted to ECFPs. Then, PCA is applied to visualize ECFP features of candidate compounds, non-candidate compounds, and actual anticancer drugs. As depicted in [Fig. 9](#), although the ECFP features of candidate compounds, non-candidate compounds, and approved anticancer drugs are distinctly separated, there is an evident overlap between the candidate compounds

and approved anticancer drugs. This suggests that the candidate compounds predicted by our model do share similar anticancer properties or features with approved anticancer drugs [60]. In contrast, this overlap is not observed in non-candidate compounds, signifying the absence of the distinctive features in the non-candidate compounds.

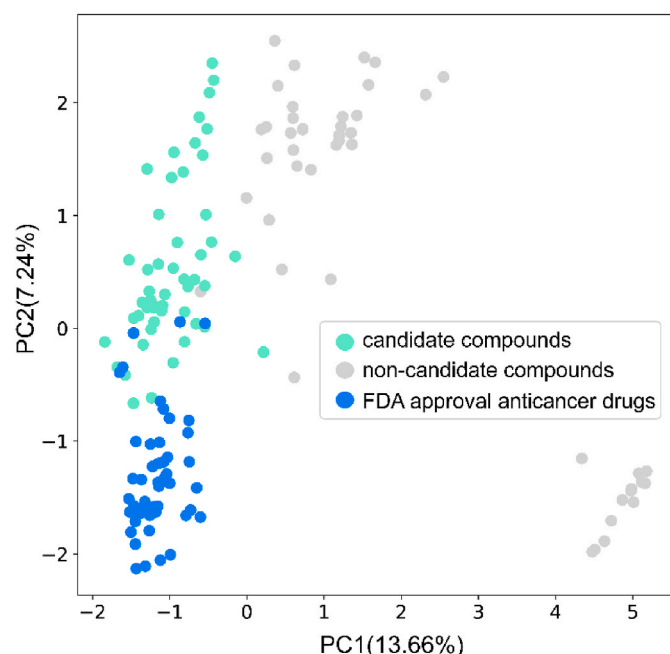
5. Discussion

Altogether, this work presents several key contributions: (i) We develop a novel cross multi-head attention oriented bidirectional interpretable CPI prediction model named CmhAttCPI. (ii) We visualized atom and residue attention weights generated from the cross multi-head attention module of CmhAttCPI and associated the high-attention residues to specific biological functions to interpret compound-protein interactions from both the chemical and biological perspective. (iii) To comprehensively evaluate the generalization ability of CmhAttCPI, we constructed three evaluation scenarios based on compound and protein clustering and evaluated model generalization performance in three

Table 3

The top ten predicted candidate compounds for breast cancer therapy.

Rank	Drug name	Target name (gene)	Evidence	
			PMID	Cell line
1	Astaxanthin	Receptor tyrosine-protein kinase erbB-2 (ERBB2, Her2)	30,287,735	MDA-MB-231
2	Voacamine	Receptor tyrosine-protein kinase erbB-2 (ERBB2, Her2)	35,278,901	MCF-7; 4T1
3	Nabiximols	Receptor tyrosine-protein kinase erbB-2 (ERBB2, Her2)	Unknown	Unknown
4	Suramin	Cellular tumor antigen p53 (TP53)	Unknown	Unknown
5	Astaxanthin	Threonine-protein kinase (AKT1)	Unknown	Unknown
6	Tetrandrine	Receptor tyrosine-protein kinase erbB-2 (ERBB2, Her2)	31,934,280	MDA-MB-231; MCF-7
7	Deacetoxyvinzolidine	Receptor tyrosine-protein kinase erbB-2 (ERBB2, Her2)	Unknown	Unknown
8	Succinobucol	Threonine-protein kinase (AKT1)	Unknown	Unknown
9	Telapristone acetate	Cadherin-1 (CDH1)	Unknown	Unknown
10	WX-UK1	Receptor tyrosine-protein kinase erbB-2 (ERBB2, Her2)	Unknown	Unknown

**Fig. 9.** ECFP feature visualization in PCA for candidate compounds, non-candidate compounds and actual anticancer drugs.

scenarios with different clustering thresholds. The results confirm the outstanding effectiveness and generalization ability of CmhAttCPI. (iv) We conducted two case studies to prove the applicability of CmhAttCPI in practices. We first applied CmhAttCPI to predict compounds with anti-breast-cancer properties. Among the predicted top ten compounds, most can be validated in vitro or published literatures. Subsequently, we utilized CmhAttCPI to discover anticancer candidate compounds, and we observed that these candidates indeed share similar anticancer features with approved anticancer drugs.

Evaluation results on four public datasets demonstrate that

CmhAttCPI consistently achieves improved effectiveness and robustness compared to other related state-of-the-art CPI deep learning models, regardless of whether the datasets are balanced or imbalanced. In addition, a series of experiments conducted across three scenarios with different clustering thresholds manifest the generalization ability of CmhAttCPI, providing a comprehensive insight into its adaptability to various conditions. Although AI-Bind outperforms CmhAttCPI in terms of average accuracy in the protein clustering unknown setting with the threshold of 0.5, CmhAttCPI achieves a significantly higher AUC than AI-Bind. Given the finding of a previous simulation experiment that AUC is a better measure than accuracy when evaluating and comparing classifiers [61], the generalization ability of CmhAttCPI is considered superior to AI-Bind. Based on these observations, CmhAttCPI has the excellent generalization ability to discover potential targets that may act on new molecules and predict new therapeutic targets and side effects for known drugs.

Furthermore, attention weight visualizations of Navitoclax-Bcl2 and BIRB 796-p38 interactions imply that CmhAttCPI is capable of providing bidirectional interpretation through focusing on important atoms and residues. This capability aids researchers in identifying unknown binding sites and providing chemical and biological interpretation for the prediction results. What's more, associating the high-attention residues with biological functions further enhance the interpretability of CmhAttCPI.

In our case studies, we firstly predicted potential anti-breast-cancer compounds among the drugs not yet approved by the FDA. Among the top ten predicted compounds, three have been validated in vitro, and four have literature confirmations of their therapeutic properties and suppression of breast cancer metastasis. Then, we applied CmhAttCPI to discover anticancer candidate compounds. The feature visualization for candidate compounds, non-candidate compounds and actual anticancer drugs reveals that the anticancer candidate compounds selected by our model indeed possess similar anticancer properties to actual anticancer drugs, while non-candidate compounds do not exhibit such properties. These findings illustrate the capability of CmhAttCPI to server as a tool for anticancer drug discovery in practices.

Although CmhAttCPI exhibits advantages in improving performance compared to existing models commonly used in CPI prediction, there is still room for further improvement in the following two perspectives. First, CmhAttCPI does not incorporate biological information, such as protein-protein interactions, drug-drug interactions, side effects, and diseases. With the significant advances in multimodal learning [62,63], we will attempt to integrate more biological data as input further enhance the prediction performance. Second, CmhAttCPI does not explore the geometric relationships between global features (the relationship between the whole CPI space and positive CPIs) and individual features (interaction features of the specific compound-protein pair). In future work, we will optimize CmhAttCPI by introducing a novel hierarchical structure, such as the Caps-Score layer, to capture geometric relationships between global features and individual features [64].

6. Conclusions

In this study, we present a novel bidirectional interpretable deep learning framework called CmhAttCPI to identify compound-protein interactions. Differing from previous models for predicting CPIs, our framework integrates two multi-head scaled dot-product attentions into the cross multi-head attention module, facilitating the efficient learning of crucial atoms and residues in CPIs. For compound feature extraction in the GCW module, we attached a graph warp unit to GCNs, increasing the representation power for molecular graphs. For protein feature extraction in the CNN module, we employed multiple parallel convolutional layers to capture residue features. Our experimental results confirm the excellent effectiveness, robustness, generalization ability and practical applicability of CmhAttCPI in predicting CPIs. While designed for CPI prediction, CmhAttCPI can also be adapted to other

tasks, such as the prediction of compound-compound interactions, protein-protein interactions, and drug-disease associations.

Data and code availability

The four public CPI datasets in this study are available at https://drive.google.com/drive/folders/1lfVZNHlpgdlBhozK_oeS1upU8NvpBOx6?usp=share_link. The implementation codes of CmhAttCPI are available at <https://github.com/wangmeng-code/CmhAttCPI>.

CRedit authorship contribution statement

Meng Wang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jianmin Wang:** Writing – review & editing, Visualization, Software. **Zhiwei Rong:** Methodology, Writing – review & editing. **Liuying Wang:** Writing – review & editing, Data curation. **Zhenyi Xu:** Writing – review & editing, Data curation. **Liu-chao Zhang:** Writing – review & editing. **Jia He:** Data curation. **Shuang Li:** Data curation. **Lei Cao:** Writing – review & editing. **Yan Hou:** Writing – review & editing, Investigation, Conceptualization. **Kang Li:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare no conflict(s) of interest.

Acknowledgments

This work has been supported by the National Natural Science Foundation of China under Grant No. 82273734 and 82173615.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compbiomed.2024.108239>.

References

- [1] S. Pushpakom, F. Iorio, P.A. Eyers, K.J. Escott, S. Hopper, A. Wells, A. Doig, T. Williams, J. Latimer, C. McNamee, Drug repurposing: progress, challenges and recommendations, *Nat. Rev. Drug Discov.* 18 (2019) 41–58.
- [2] E. Gregori-Puigjané, V. Setola, J. Hert, B.A. Crews, J.J. Irwin, E. Lounkine, L. Marnett, B.L. Roth, B.K. Shoichet, Identifying mechanism-of-action targets for drugs and probes, *Proc. Natl. Acad. Sci. USA* 109 (2012) 11178–11183.
- [3] S.M. Paul, D.S. Mytelka, C.T. Dunwiddie, C.C. Persinger, B.H. Munos, S. R. Lindborg, A.L. Schacht, How to improve R&D productivity: the pharmaceutical industry's grand challenge, *Nat. Rev. Drug Discov.* 9 (2010) 203–214.
- [4] M. Bagherian, E. Sabeti, K. Wang, M.A. Sartor, Z. Nikolovska-Coleska, K. Najarian, Machine learning approaches and databases for prediction of drug–target interaction: a survey paper, *Briefings Bioinf.* 22 (2021) 247–269.
- [5] I. Lee, J. Keum, H. Nam, DeepConv-DTI: prediction of drug–target interactions via deep learning with convolution on protein sequences, *PLoS Comput. Biol.* 15 (2019) e1007129.
- [6] M. Hamanaka, K. Taneishi, H. Iwata, J. Ye, J. Pei, J. Hou, Y. Okuno, CGBVS-DNN: prediction of compound–protein interactions based on deep learning, *Molecular informatics* 36 (2017) 1600045.
- [7] M. Wen, Z. Zhang, S. Niu, H. Sha, R. Yang, Y. Yun, H. Lu, Deep-learning-based drug–target interaction prediction, *J. Proteome Res.* 16 (2017) 1401–1409.
- [8] M. Tsubaki, K. Tomii, J. Sese, Compound–protein interaction prediction with end-to-end learning of neural networks for graphs and sequences, *Bioinformatics* 35 (2019) 309–318.
- [9] J. Wang, X. Li, H. Zhang, GNN-PT: Enhanced Prediction of Compound–Protein Interactions by Integrating Protein Transformer, 2020 arXiv preprint arXiv: 2009.00805.
- [10] T. Cai, L. Xie, S. Zhang, M. Chen, D. He, A. Badkul, Y. Liu, H.K. Namballa, M. Dorogan, W.W. Harding, End-to-end sequence–structure–function meta-learning predicts genome-wide chemical–protein interactions for dark proteins, *PLoS Comput. Biol.* 19 (2023) e1010851.
- [11] H. Öztürk, A. Özgür, E. Ozkirimli, DeepDTA: deep drug–target binding affinity prediction, *Bioinformatics* 34 (2018) i821–i829.
- [12] W. Torng, R.B. Altman, Graph convolutional neural networks for predicting drug–target interactions, *J. Chem. Inf. Model.* 59 (2019) 4131–4149.
- [13] M. Karimi, D. Wu, Z. Wang, Y. Shen, DeepAffinity: interpretable deep learning of compound–protein affinity through unified recurrent and convolutional neural networks, *Bioinformatics* 35 (2019) 3329–3338.
- [14] D. Weininger, SMILES, a chemical language and information system. 1. Introduction to methodology and encoding rules, *J. Chem. Inf. Comput. Sci.* 28 (1988) 31–36.
- [15] T. Nguyen, H. Le, T.P. Quinn, T. Nguyen, T.D. Le, S. Venkatesh, GraphDTA: predicting drug–target binding affinity with graph neural networks, *Bioinformatics* 37 (2021) 1140–1147.
- [16] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need, *Adv. Neural Inf. Process. Syst.* (2017) 30.
- [17] L. Chen, X. Tan, D. Wang, F. Zhong, X. Liu, T. Yang, X. Luo, K. Chen, H. Jiang, M. Zheng, TransformerCPI: improving compound–protein interaction prediction by sequence-based deep learning with self-attention mechanism and label reversal experiments, *Bioinformatics* 36 (2020) 4406–4414.
- [18] L. Huang, J. Lin, R. Liu, Z. Zheng, L. Meng, X. Chen, X. Li, K.-C. Wong, CoaDTI: multi-modal co-attention based framework for drug–target interaction annotation, *Briefings Bioinf.* 23 (2022) bbac446.
- [19] H. Kurata, S. Tsukiyama, ICAN: interpretable cross-attention network for identifying drug and target protein interactions, *PLoS One* 17 (2022) e0276609.
- [20] A. Chatterjee, R. Walters, Z. Shafi, O.S. Ahmed, M. Sebek, D. Gysi, R. Yu, T. Eliassi-Rad, A.-L. Barabási, G. Menichetti, Improving the generalizability of protein–ligand binding predictions with AI-Bind, *Nat. Commun.* 14 (2023) 1989.
- [21] Q. Ye, C.-Y. Hsieh, Z. Yang, Y. Kang, J. Chen, D. Cao, S. He, T. Hou, A unified drug–target interaction prediction framework based on knowledge graph and recommendation system, *Nat. Commun.* 12 (2021) 6775.
- [22] R.P. Sheridan, Time-split cross-validation as a method for estimating the goodness of prospective prediction, *J. Chem. Inf. Model.* 53 (2013) 783–790.
- [23] A. Mayr, G. Klambauer, T. Unterthiner, M. Steijaert, J.K. Wegner, H. Ceulemans, D.-A. Clevert, S. Hochreiter, Large-scale comparison of machine learning methods for drug target prediction on ChEMBL, *Chem. Sci.* 9 (2018) 5441–5451.
- [24] M.K. Gilson, T. Liu, M. Baitaluk, G. Nicola, L. Hwang, J. Chong, BindingDB in 2015: a public database for medicinal chemistry, computational chemistry and systems pharmacology, *Nucleic Acids Res.* 44 (2016) D1045–D1053.
- [25] D.S. Wishart, C. Knox, A.C. Guo, D. Cheng, S. Shrivastava, D. Tzur, B. Gautam, M. Hassanali, DrugBank: a knowledgebase for drugs, drug actions and drug targets, *Nucleic Acids Res.* 36 (2008) D901–D906.
- [26] M. Zitnik, M. Agrawal, J. Leskovec, Modeling polypharmacy side effects with graph convolutional networks, *Bioinformatics* 34 (2018) i457–i466.
- [27] W.K.B. Chan, H. Zhang, J. Yang, J.R. Brender, J. Hur, A. Özgür, Y. Zhang, GLASS: a comprehensive database for experimentally validated GPCR–ligand associations, *Bioinformatics* 31 (2015) 3035–3042.
- [28] S. Ryu, Y. Kwon, W.Y. Kim, A Bayesian graph convolutional network for reliable prediction of molecular properties with uncertainty quantification, *Chem. Sci.* 10 (2019) 8438–8446.
- [29] Z. Wu, D. Jiang, C.-Y. Hsieh, G. Chen, B. Liao, D. Cao, T. Hou, Hyperbolic relational graph convolution networks plus: a simple but highly efficient QSAR-modeling method, *Briefings Bioinf.* 22 (2021) bbab112.
- [30] J. Li, D. Cai, X. He, Learning Graph-Level Representation for Drug Discovery, 2017 arXiv preprint arXiv:1709.03741.
- [31] C. Shang, Q. Liu, K.-S. Chen, J. Sun, J. Lu, J. Yi, J. Bi, Edge Attention-Based Multi-Relational Graph Convolutional Networks, 2018 arXiv preprint arXiv: 1802.04944.
- [32] K. Ishiguro, S.-i. Maeda, M. Koyama, Graph Warp Module: an Auxiliary Module for Boosting the Power of Graph Neural Networks in Molecular Graph Analysis, 2019 arXiv preprint arXiv:1902.01020.
- [33] W. Rawat, Z. Wang, Deep convolutional neural networks for image classification: a comprehensive review, *Neural Comput.* 29 (2017) 2352–2449.
- [34] J. Peng, J. Li, X. Shang, A learning-based method for drug–target interaction prediction based on feature representation learning and deep neural network, *BMC Bioinf.* 21 (2020) 1–13.
- [35] D. Bahdanau, K. Cho, Y. Bengio, Neural Machine Translation by Jointly Learning to Align and Translate, 2014 arXiv preprint arXiv:1409.0473.
- [36] C.d. Santos, M. Tan, B. Xiang, B. Zhou, Attentive Pooling Networks, 2016 arXiv preprint arXiv:1602.03609.
- [37] J.C. Gower, G.J.S. Ross, Minimum spanning trees and single linkage cluster analysis, *J. Roy. Stat. Soc.: Series C (Applied Statistics)* 18 (1969) 54–64.
- [38] G. Landrum, RDKit: Open-Source Cheminformatics, 2006.
- [39] T.T. Ashburn, K.B. Thor, Drug repositioning: identifying and developing new uses for existing drugs, *Nat. Rev. Drug Discov.* 3 (2004) 673–683.
- [40] A.S. Reddy, S. Zhang, Polypharmacology: drug discovery for the future, *Expert Rev. Clin. Pharmacol.* 6 (2013) 41–47.
- [41] E. Lounkine, M.J. Keiser, S. Whitebread, D. Mikhailov, J. Hamon, J.L. Jenkins, P. Lavan, E. Weber, A.K. Doak, S. Côté, Large-scale prediction and testing of drug activity on side-effect targets, *Nature* 486 (2012) 361–367.
- [42] M. Schenone, V. Dančák, B.K. Wagner, P.A. Clemons, Target identification and mechanism of action in chemical biology and drug discovery, *Nat. Chem. Biol.* 9 (2013) 232–240.
- [43] K.W. Yip, J.C. Reed, Bcl-2 family proteins and cancer, *Oncogene* 27 (2008) 6398–6406.
- [44] Z. Huang, Bcl-2 family proteins as targets for anticancer drug design, *Oncogene* 19 (2000) 6627–6631.
- [45] J. Zwerina, S. Hayer, K. Redlich, K. Bobacz, G. Kollias, J.S. Smolen, G. Schett, Activation of p38 MAPK is a key step in tumor necrosis factor-mediated inflammatory bone destruction, *Arthritis Rheum.: Official Journal of the American College of Rheumatology* 54 (2006) 463–472.

- [46] A.J. Souers, J.D. Levenson, E.R. Boghaert, S.L. Ackler, N.D. Catron, J. Chen, B. D. Dayton, H. Ding, S.H. Enschede, W.J. Fairbrother, ABT-199, a potent and selective BCL-2 inhibitor, achieves antitumor activity while sparing platelets, *Nat. Med.* 19 (2013) 202–208.
- [47] C. Pargellis, L. Tong, L. Churchill, P.F. Cirillo, T. Gilmore, A.G. Graham, P.M. Grob, E.R. Hickey, N. Moss, S. Pav, Inhibition of p38 MAP kinase by utilizing a novel allosteric binding site, *Nat. Struct. Biol.* 9 (2002) 268–272.
- [48] T.W. Sedlak, Z.N. Oltvai, E. Yang, K. Wang, L.H. Boise, C.B. Thompson, S. J. Korsmeyer, Multiple Bcl-2 family members demonstrate selective dimerizations with Bax, *Proc. Natl. Acad. Sci. USA* 92 (1995) 7834–7838.
- [49] X.-M. Yin, Z.N. Oltvai, S.J. Korsmeyer, BH1 and BH2 domains of Bcl-2 are required for inhibition of apoptosis and heterodimerization with Bax, *Nature* 369 (1994) 321–323.
- [50] A.A. Ivanov, F.R. Khuri, H. Fu, Targeting protein–protein interactions as an anticancer strategy, *Trends Pharmacol. Sci.* 34 (2013) 393–400.
- [51] A. Letai, BH3 domains as BCL-2 inhibitors: prototype cancer therapeutics, *Expert Opin. Biol. Ther.* 3 (2003) 293–304.
- [52] A. Jemal, F. Bray, M.M. Center, J. Ferlay, E. Ward, D. Forman, Global cancer statistics, *CA A Cancer J. Clin.* 61 (2011) 69–90.
- [53] A.G. Waks, E.P. Winer, Breast cancer treatment: a review, *JAMA* 321 (2019) 288–300.
- [54] M.H. Bailey, C. Tokheim, E. Porta-Pardo, S. Sengupta, D. Bertrand, A. Weerasinghe, A. Colaprico, M.C. Wendl, J. Kim, B. Reardon, Comprehensive characterization of cancer driver genes and mutations, *Cell* 173 (2018) 371–385.
- [55] B. McCall, C.K. McPartland, R. Moore, A. Frank-Kamenetskii, B.W. Booth, Effects of astaxanthin on the proliferation and migration of breast cancer cells in vitro, *Antioxidants* 7 (2018) 135.
- [56] Y. Zuo, C.-z. Zhang, Q. Ren, Y. Chen, X. Li, J.-r. Yang, H.-x. Li, W.-t. Tang, H. O. Hing-Man, C. Sun, Activation of mitochondrial-associated apoptosis signaling pathway and inhibition of PI3K/Akt/mTOR signaling pathway by voacamine suppress breast cancer progression, *Phytomedicine* 99 (2022) 154015.
- [57] B. Yu, B. Yuan, A. Kiyomi, H. Kikuchi, H. Hayashi, X. Hu, M. Okazaki, M. Sugiura, T. Hirano, X. Pei, Differentiation induction of human breast cancer cells by arsenite in combination with tetrandrine, *Am. J. Tourism Res.* 11 (2019) 7310.
- [58] D. Rogers, M. Hahn, Extended-connectivity fingerprints, *J. Chem. Inf. Model.* 50 (2010) 742–754.
- [59] S. Wold, K. Esbensen, P. Geladi, Principal component analysis, *Chemometr. Intell. Lab. Syst.* 2 (1987) 37–52.
- [60] J. Wang, J. Mao, M. Wang, X. Le, Y. Wang, Explore drug-like space with deep generative models, *Methods* 210 (2023) 52–59.
- [61] C.X. Ling, J. Huang, H. Zhang, AUC: a Better Measure than Accuracy in Comparing Learning Algorithms, *Springer*, pp. 329–341.
- [62] A. Dehghan, P. Razzaghi, K. Abbasi, S. Gharaghani, TripletMultiDTI: Multimodal Representation Learning in Drug-Target Interaction Prediction with Triplet Loss Function, *Expert systems with applications*, 2023 120754.
- [63] F. Rafiei, H. Zeraati, K. Abbasi, J.B. Ghasemi, M. Parsaeian, A. Masoudi-Nejad, DeepTraSynergy: drug combinations using multimodal deep learning with transformers, *Bioinformatics* 39 (2023) btad438.
- [64] K. Abbasi, P. Razzaghi, Incorporating part-whole hierarchies into fully convolutional network for scene parsing, *Expert Syst. Appl.* 160 (2020) 113662.